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Model Selection, Confounder Control, and Marginal Structural Models: Review and New Applications

Marshall M. JOFFE, Thomas R. TEN HAVE, Harold I. FELDMAN, and Stephen E. KIMMEL

In traditional regression modeling, to control for confounding by a variable one must include it in the structural part of the statistical model. Marginal structural models are a flexible new set of causal models. The estimation methods used to estimate model parameters use weighting to control for confounding; this allows more flexibility in choosing covariates for inclusion in the structural model and allows the model to more precisely reflect the scientific questions of interest. An important example of this is in multicenter observational studies where there is confounding by cluster. We illustrate these points with data from a study of surgery to provide vascular access for hemodialysis and a study comparing different timings for coronary angioplasty.

KEY WORDS: Causality; Prediction; Weighting.

1. INTRODUCTION

One of the main uses of regression models has been to estimate the effect of a treatment or exposure on some outcome of interest. This use has been widespread, although not without controversy (Freedman 1999). In standard regression modeling, controlling for confounding by a variable requires including it in the structural part of a statistical model. This is unfortunate, because these variables may be mere nuisance variables of little wider interest. A new class of causal models, the marginal structural models (MSMs), and their associated weighted estimation (Hernan, Brumback, and Robins 2000, 2001, 2002; Robins 1999; Robins, Hernan, and Brumback 2000) allow separation of model selection from confounder control, so permitting one to keep variables of little intrinsic interest out of the structural part of the model while still controlling for confounding by those variables. Thus, these models provide a useful addition to standard regression methods. One promising use is for control of confounding by cluster in multicenter observational studies, where standard methods (Liang and Zeger 1986) for estimating marginal models (i.e., that do not include cluster in the structural model) and mixed effects models are inappropriate (Berlin, Kimmel, Ten Have, and Sammel 1999). We illustrate

these points with data from two studies: a study of predictors of successful placement of an arteriovenous fistula (AVF) for subjects requiring hemodialysis illustrates the models' general usefulness, and a study comparing two methods for performing percutaneous transluminal coronary angioplasty (PTCA) deals specifically with confounding by cluster.

2. CAUSAL ESTIMANDS AND MARGINAL STRUCTURAL MODELS

2.1 The Potential Outcomes Model

The potential outcomes approach to causality defines the effect of an intervention (e.g., treatment or exposure) as a comparison of what would happen to some individual or group under different treatment or exposure conditions (Rubin 1974). Before introducing a model for causal effects, we require notation for observed and latent quantities. Let i denote the unit of observation or subject. Let A_i denote the observed level of the treatment whose effect we wish to measure; the subscript i is to be understood when referring to variables but is usually suppressed. We use the lowercase a to denote a hypothetical level of treatment. Let Y denote the outcome of interest. Y may be a scalar or vector; its components may be continuous or discrete. Let $\mathbf{X}$ denote measured pretreatment covariates; in multicenter studies, $\mathbf{X}$ can include the center or cluster indicator. Let $\mathbf{V} \in \mathbf{X}$ denote specified pretreatment covariates whose associations with outcome are of interest. All elements of $\mathbf{V}$ are in $\mathbf{X}$, but the converse will not be true; we may classify those elements in $\mathbf{X}$ but not in $\mathbf{V}$ as nuisance variables. We discuss later the usefulness of distinguishing among pretreatment covariates.

Let Y_i^a denote the outcome Y that we would see in subject i were that subject to receive level a of treatment. We call Y^a a potential outcome because, until the time a decision is made about treatment, the outcome Y^a observed under any given treatment a is potentially observable. If a subject receives treatment level a (i.e., if $A = a$), Y^a is observed; otherwise, Y^a is not observed and is counterfactual. Let $\mathbf{Y} \equiv \{Y^a\}$ denote the vector of all potential outcomes Y^a for a given subject i .

Causal effects are comparisons of different potential outcomes Y_i^a for the same units i under different treatment levels a and a' . One cannot directly compare potential outcomes Y_i^a and $Y_i^{a'}$, $a \neq a'$ for any individual subject, because the outcomes Y_i^a and $Y_i^{a'}$ are not simultaneously observable in subject i . However, one can estimate the distributions or expectations of potential outcomes in a group defined by pretreatment covariates $\mathbf{V}$. Let $f(\mathbf{Y}|\mathbf{V})$ denote the joint density of the potential outcomes $\mathbf{Y} \equiv \{Y^a\}$ in a subset of the data defined by covariates $\mathbf{V}$, and let $f(Y^a|\mathbf{V})$ denote the density of the potential outcome Y^a in that subset; we call $f(Y^a|\mathbf{V})$ the marginal density, because it is marginal with respect to other potential outcomes $Y^{a'}$, $a' \neq a$. The joint density is not generally estimable,

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even from experimental data; however, the marginal densities (or their corresponding expectations) are estimable under assumptions that are often made (Section 2.3). Comparisons of the marginal densities $f(Y^a|\mathbf{V})$ and $f(Y^{a'}|\mathbf{V})$, $a \neq a'$, represent the causal effects of treatment on the aggregate or group defined by pretreatment covariates $\mathbf{V}$.

2.2 Marginal Structural Models

Marginal structural models (MSMs) provide a way to model these aggregate causal effects (Joffe 2001). MSMs are models for the marginal densities or expectations of the potential outcomes Y^a given specified baseline covariates $\mathbf{V}$; that is, they are models for $f(Y^a|\mathbf{V})$ (densities) or $E(Y^a|\mathbf{V})$ (expectations); we concentrate on MSMs for expectations. The MSM expectation model takes the mathematical form

$$E(Y^a|\mathbf{V}) = q(a, \mathbf{V}); \quad (1)$$

that is, the expectation of the outcome that would be seen under treatment level a for subjects with covariates $\mathbf{V}$ is a possibly unknown function $q(a, \mathbf{V})$ of the treatment level a to be applied and the covariates $\mathbf{V}$.

Typically, a known form will be picked for $q(\cdot)$; for example, for a binary outcome, one might let

$$E(Y^a|\mathbf{V}) = q(a, \mathbf{V}) = \text{expit}(a\beta_a + \mathbf{V}\beta_V), \quad (2)$$

where β_a and β_V are (possibly vector) regression parameters corresponding to a and $\mathbf{V}$, and $\text{expit}(r) \equiv \{1 + \exp(-r)\}^{-1}$ is the inverse of the logit transform; that is, $\text{logit}\{\text{expit}(r)\} = r$. In Equation (2), the effect of treatment on the odds of success is the same across all levels of covariates $\mathbf{V}$, and the associations of covariates $\mathbf{V}$ with the odds of success under a specified treatment level a are the same for all a .

2.3 Additional Restrictions Allowing Estimation

To estimate the parameters of MSMs and so estimate causal effects, we require additional assumptions. First, we restrict attention to discrete (rather than continuous) treatments or exposures. In addition, we assume that the potential outcomes $\mathbf{Y}$ do not predict treatment assignment once one accounts for baseline covariates $\mathbf{X}$, and that there is a nonzero probability that subjects with any covariate level $\mathbf{X}$ receive treatment level a ; that is,

$$\Pr(A = a|\mathbf{X}, \mathbf{Y}) = \Pr(A = a|\mathbf{X}) > 0 \quad (3)$$

for all $a, \mathbf{X}$. Condition (3) is known as strong ignorability of treatment assignment (Rosenbaum and Rubin 1983); equivalently, within strata defined by covariates $\mathbf{X}$, groups assigned to various levels of treatment are comparable except for the effect of treatment (Greenland and Robins 1986).

3. ESTIMATION OF MARGINAL STRUCTURAL MODELS

Under ignorability and our modeling assumptions, we use the following steps to estimate the parameters in our MSM:

1. Model the association of covariates and treatment assignment.
2. Use this model to derive weights.
3. Use the weights to create an artificial population in which covariates are not associated with treatment.

4. Estimate the parameters in our MSM using weighted methods.

We illustrate the steps in our estimation procedure using data from a study of predictors of successful placement of an arteriovenous fistula (AVF) for subjects requiring hemodialysis. Before discussing these steps, we provide some background about AVF surgery and our study.

3.1 A Study of Arteriovenous Fistula Surgery

Subjects with end stage renal disease (ESRD) require dialysis or a kidney transplant to replace the lack of function of their failed native kidneys. Hemodialysis is the most common form of kidney replacement, and its successful implementation depends on the ability to maintain a surgically created vascular conduit capable of sustaining high blood flows. A surgically created native arteriovenous fistula (AVF) is one such conduit, and has several advantages over alternative approaches, such as those involving use of synthetic vascular grafts. AVFs are created by anastomosing an artery directly to a vein, more often in the upper extremity. We say that surgery was successful ($Y = 1$) if it led to successful use of the AVF in hemodialysis and unsuccessful ($Y = 0$) if it did not. AVF surgery is less often successful in the short-term than surgery to create conduits through implantation of synthetic material. Thus, it would be desirable to know in advance whose AVF surgery is likely to succeed, and so perform the surgery only for these people.

To answer this question, we used data from a study of 348 subjects undergoing AVF surgery (Feldman et al. 2003). Answering the question is complicated by the fact that the study collected information not only on such personal characteristics $\mathbf{X}$ as age and health status, but also on characteristics of the surgical procedure itself, such as heparin use (A). Heparin is used to prevent blood clots from forming during or after surgery, which could affect the successful maturation of the graft; potential benefits of heparin were not, however, confirmed at the time of the study. Thus, we wanted to use data from the study to estimate the effect of heparin. If heparin use were found to affect the probability of success of the procedure, we would want to estimate the probability of success were someone to be provided the optimal heparin dose. We dichotomize heparin use as high-dose (at least 1,000 units during surgery, $A = 1$) or not ($A = 0$).

In determining who is a good candidate for surgery, we want to predict who would do well if provided the optimal dose of heparin. We are interested in identifying subjects whose probability of success, based on easily measurable presurgical variables $\mathbf{V}$, is high; that is, those subjects with large $E(Y^a = 1|\mathbf{V})$ is high for whichever treatment level a is deemed to be preferable. Although most covariates $\mathbf{X}$ (e.g., age, history of cardiovascular disease) are easily measured prior to surgery, the size of the vessels used in the surgical procedure is not. Thus, it is not useful in predicting the probability of surgical success for a surgeon or physician contemplating surgery.

3.2 Weighting to Remove Covariate Imbalances

To see whether the effect of heparin is likely to be confounded, we examined distributions of covariates at different levels of

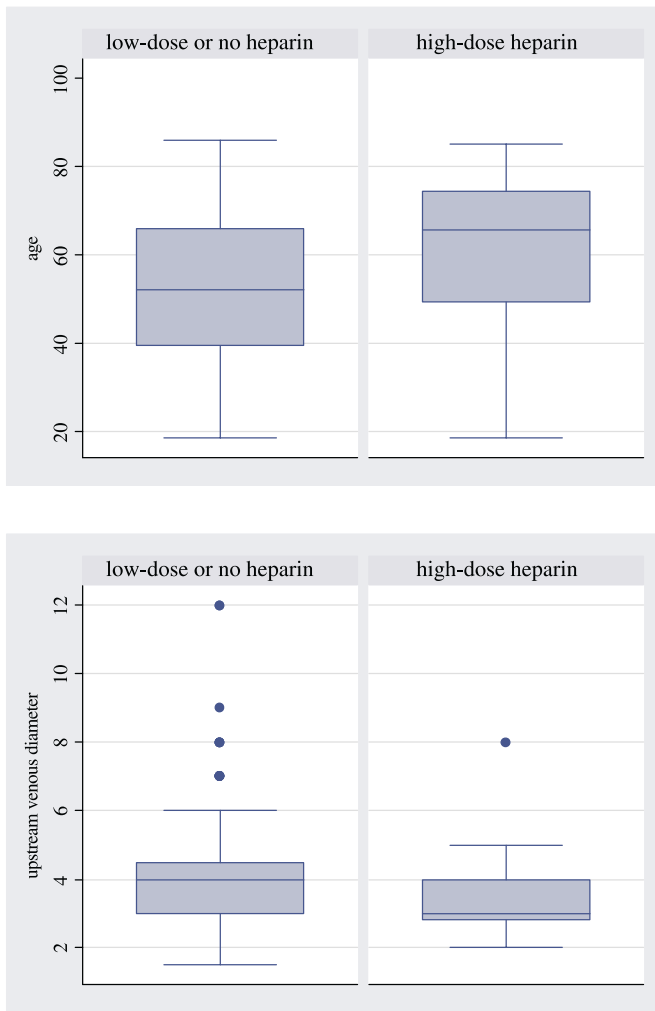


Figure 1. Boxplots of distributions of continuous covariates by heparin dose.

heparin dose. Figure 1 shows boxplots of age and vessel size among subjects receiving and not receiving high-dose heparin; subjects receiving high-dose heparin are more likely to be older and have smaller vessels used in the surgery. Table 1 summarizes the distributions of the other covariates by heparin use category. Because older age and smaller vessels are associated with poorer outcomes, one would expect analyses which do not control for these variables to understate the benefits of heparin.

We use a weighting procedure to remove these imbalances and then estimate heparin effects and predict outcomes. The weighting procedure begins by using multivariable logistic regression

to model the probability of high-dose heparin use; that is, we estimated parameters β in the model $\logit\{\Pr(A = 1|\mathbf{X})\} = \mathbf{X}\beta$ using maximum likelihood. Table 2 shows the estimates $\hat{\beta}$ of the regression coefficients. These coefficients from this model reflect the associations shown graphically before. Even after controlling for other covariates, smaller vessels are associated with high-dose heparin use; thus, we may need to control for vessel size, even though we do not want to use it in prediction.

We used the estimated coefficients to estimate weights for our subsequent analysis. Let $p_X \equiv \Pr(A = 1|\mathbf{X})$ denote a subject's probability, given covariates, of receiving treatment (high-dose heparin); p_X is known as the propensity score (Rosenbaum and Rubin 1983). If we knew the true propensity score, we could compute weights $w_{X,A}$ for our analysis as $w_{X,A} = A/p_X + (1 - A)/(1 - p_X)$; w_x is the inverse of the probability that a subject with covariates $\mathbf{X}$ received his observed treatment. Because we do not know the true weights we estimate them from our data; the estimated probability of heparin use is $\hat{p}_X \equiv \text{expit}(\mathbf{X}\hat{\beta})$, and the estimated weight is then

$$\hat{w}_{X,A} = A/\hat{p}_X + (1 - A)/(1 - \hat{p}_X).$$

We apply weights $\hat{w}_{X,A}(w_{X,A})$ to each subject in the population; each subject then represents a total of $\hat{w}_{X,A}(w_{X,A})$ subjects. Applying these weights to the study subjects creates an artificial population which tends to balance all covariates $\mathbf{X}$ used to derive the weights. We see this in heparin study: Figure 2 shows weighted boxplots of age and vessel size, and Table 1 shows distributions of categorical covariates by heparin use. The weighting balances the most egregious disparities in the covariates. In this artificial population, there can be little confounding of the effect of heparin by these covariates, since the distribution of covariates is similar at both levels of heparin use (Rosenbaum 1987).

3.3 Estimation

In contrast to standard approaches, in the artificial population created by our weighting procedure, there is no association between vessel size and heparin use, and so standard modeling procedures may be applied in this population. In practice, this involves maximizing a pseudo (log-) likelihood, where each subject's contribution is weighted by $\hat{w}_{X,A}$; for a binary outcome, the contribution of an individual with covariates $\mathbf{X}$ and treatment A is $\hat{w}_{X,A}Y * \ln\{E(Y^{A=1}|\mathbf{V}; \beta)\} + \hat{w}_{X,A}(1 - Y) * \ln\{1 - E(Y^{A=1}|\mathbf{V}; \beta)\}$. Point estimates may be obtained using regression procedures which allow weighting. The variance of this estimator may be estimated with standard estimating equation methods, using the robust variance estimator (Liang and

Table 1. Proportion of Subjects With Various Characteristics, by Heparin Dose

High-dose heparin use:	Unweighted analysis		Weighted analysis	
	No	Yes	No	Yes
Characteristic				
Low blood pressure at admission	.112	.134	.121	.147
Dependent on dialysis	.748	.680	.734	.761
Cardiovascular disease	.199	.289	.238	.282
Prior permanent access	.272	.309	.291	.349
N	206	97		

Table 2. Logistic Regression Model for Associations of Various Predictors With High-Dose Heparin Use

	Coefficient (β)	Standard error	Odds ratio	p value
Intercept	.87	.52		
Age (reference: 70+)				<.0001
≤40	−1.72	.34	.18	
41–56	−1.54	.48	.21	
57–69	−.68	.38	.51	
Prior permanent access?				.01
Yes	.67	.27	1.94	
Upstream vein diameter (reference: < 3 mm)				.04
3–4	−.46	.59	.63	
4–5	−.60	.64	.55	
>5	−1.27	.67	.28	
Dependent on dialysis?				.23
Yes	−.42	.35	.66	
Cardiovascular disease?				.57
Yes	−.10	.18	.90	
Blood pressure at admission				.73
Low (versus normal)	−.13	.38	.87	

Zeger 1986); in practice, this may be done in Stata using the robust option available in many modeling commands.

We fit a logistic MSM (2) to the AVF data; Table 3 shows the estimated odds ratios. We estimate that high-dose heparin increases the odds of successful maturation by 2.9 times (95% confidence interval (CI): 2.1 to 4.0). The associations of other variables with AVF maturation are also shown; younger age, not being on dialysis, not having cardiovascular disease, and having had previous permanent access are all associated with higher probabilities of success. Figure 3 shows the distribution of predicted outcomes under high-dose heparin in this group; there is substantial variation among subjects, and the model discriminates between subjects with high probabilities of success and those less likely to succeed (Harrell, Lee, and Mark 1996; Joffe, Knauss, Rosas, and Feldman 2003).

Figure 3 shows the predicted probabilities $E(Y^a|\mathbf{V})$ of successful AVF maturation if given high-dose heparin, the treatment level that appears superior according to our analysis. There is substantial spread in these values, and so the model appears useful for discriminating between subjects whose surgery is likely to succeed if high-dose heparin is provided and those whose surgery is unlikely to succeed. Formal methods for evaluating the utility of the model for discrimination are discussed elsewhere (Joffe et al. 2003).

We contrast this model with several other models (Table 4). We consider three conventional logistic models, estimated without weighting, and corresponding weighted models. These include models controlling for

1. no other covariates;
2. the same covariates $\mathbf{V}$ as are in our logistic MSM; and
3. all covariates controlled through weighting.

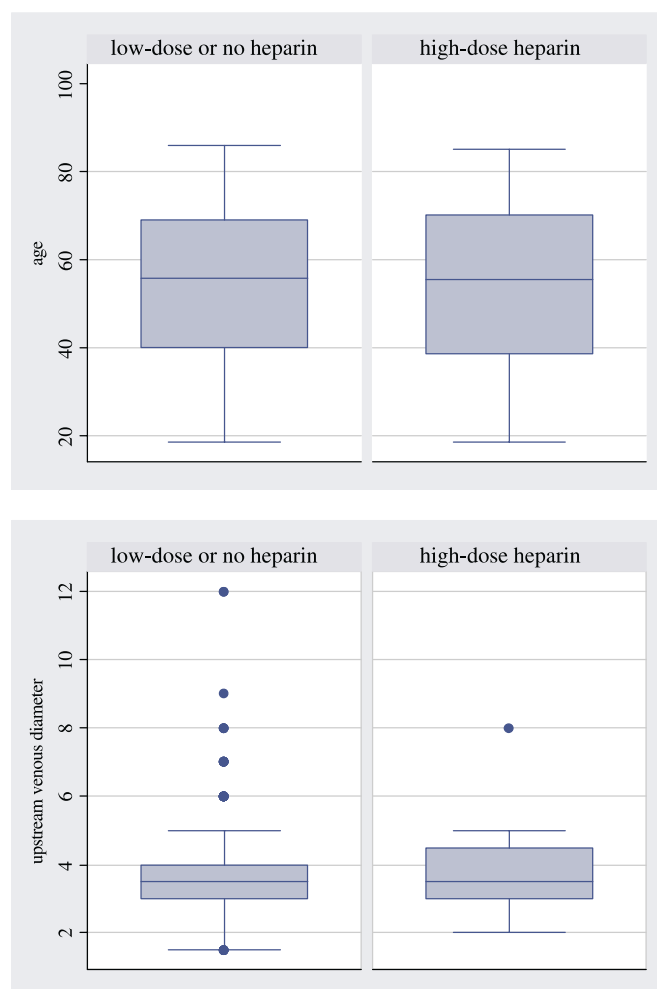


Figure 2. Weighted boxplots of distributions of continuous covariates by heparin dose.

Table 3. Marginal Structural Logistic Regression Model for Successful AVF Maturation

	Coefficient (β)	Standard error	Odds ratio	p value
Intercept	.26	.45		
High-dose heparin Yes	1.05	.17	2.86	<.0001
Age (reference: 70+)				.075
≤40	1.15	.52	3.15	
41–56	.63	.40	1.87	
57–69	.10	.25	1.11	
Prior permanent access? Yes	.48	.23	1.61	.035
Dependent on dialysis? Yes	–.68	.31	.51	.028
Cardiovascular disease? Yes	–1.24	.38	.29	.001
Blood pressure at admission Low (versus normal)	–.45	.38	.64	.250

Although Model 2 has the same form as our MSM, the odds ratio for high-dose heparin was 2.2. The lower odds ratio reflects that the conventional model does not control for confounding by vessel size; in contrast, our MSM estimates control confounding by vessel size through weighting. Similarly, the effect of heparin in unweighted and weighted forms of Model 1, having no covariates in the structural model, differ substantially. Although the weighted form of Model 1 provides a valid estimate of a causal odds ratio, it is useless for identifying subjects whose surgery is likely to succeed.

By contrast, when vessel size is included in the structural model, unweighted and weighted estimates of heparin effect are similar (odds ratios of 2.7 and 2.9, respectively), because both models control for confounding by vessel size. These models for $E(Y|\mathbf{X}, A)$ or $E(Y^a|\mathbf{X})$ do not, however, directly make the predictions $E(Y^a|\mathbf{V})$ of greatest clinical interest. We can, however, use them indirectly to estimate $E(Y^a|\mathbf{V})$ as $\sum_{\mathbf{X}} \Pr(\mathbf{X}|\mathbf{V})E(Y^a|\mathbf{X}, \mathbf{V})$, using model-based estimates of $E(Y^a|\mathbf{X}, \mathbf{V})$ (Flanders and Rhodes 1987; Joffe and Greenland 1995). This is more difficult to implement than our current approach.

4. CONFOUNDING BY CENTER: A STUDY OF CORONARY ANGIOPLASTY

Many studies include data from multiple centers. It is well understood that account must be made of center for correct variance calculations, but less well understood that center may confound the effect of the treatment or exposure of interest (Berlin et al. 1999; Localio, Berlin, Ten Have, and Kimmel 2001). To estimate the variance correctly, accepted approaches include

1. conditional inference (e.g., conditional logistic regression, the stratified proportional hazards model);
2. mixed effects models; and

3. generalized estimating equations for estimation of parameters in marginal models.

Of these approaches, only conditional inference correctly deals with confounding by center or cluster. One can also correctly use

4. standard fixed effects models if the cluster is entered as a fixed effect

although performance will be poor under the logit link if the number of strata is large (Breslow and Day 1980).

To explain why approaches 2 and 3 may not control appropriately for confounding by cluster, note that, under ignorable treatment assignment (3), subjects within a given center are comparable, except for the effects of the treatment or exposure of interest. This implies that comparisons of treated and untreated subjects within a given center (i.e., within-center effects) are valid measures of causal effects. When center is associated with treatment and a single variable for treatment is entered in the model, GEE and mixed effects models produce coefficients that represent different combinations of the within-center effect and of the association of the average treatment within center and outcome (Berlin et al. 1999; Chao, Palta, and Young 1997; Neuhaus and Kalbfleisch 1998; Pepe and Anderson 1994); this latter association, under our assumptions (3), does not represent the causal effect of treatment.

Both approaches that control confounding by cluster (1 and 4) produce inference conditional on cluster. Sometimes, marginal inference is desirable; MSMs generalize marginal models and the associated estimation methods to account for confounding by cluster and the nonindependence of subjects within cluster.

We illustrate this with data from a large study of major complications resulting from different timing of percutaneous coronary angioplasty (PTCA), a mechanical means of relieving blockage of coronary blood vessels (Kimmel et al. 1997), based on the

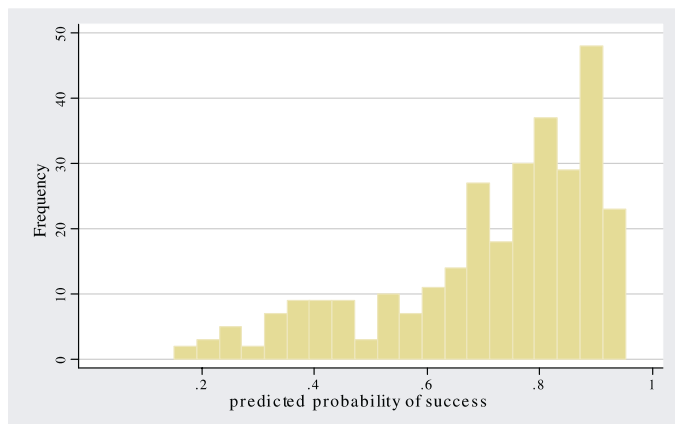


Figure 3. Histogram of predicted probabilities of success, $E(Y^1|X; \hat{\beta})$ in AVF study.

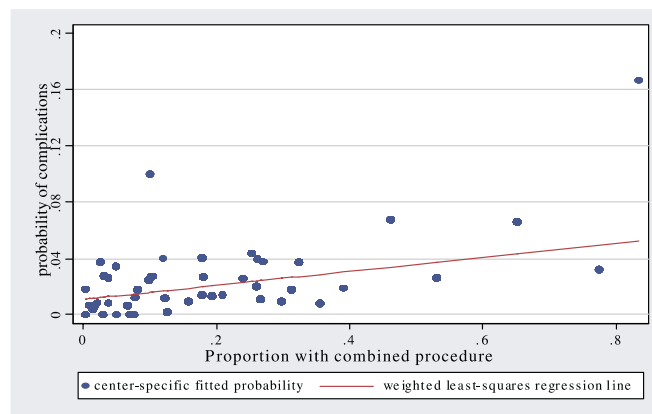


Figure 4. Center-level associations of Procedure type and probability of complications.

1992 and 1993 Registries of the Society for Cardiac Angiography. Under the staged procedure, PTCA is performed at a separate time after initial diagnostic catheterization. Under the combined procedure, PTCA is performed at the same time as the diagnostic catheterization. This study included data from 49 centers, with a total of 18,568 subjects. For all our analyses, we excluded the two smallest centers, with 1 and 2 subjects, respectively; their inclusion in the analysis would result in estimated probabilities of the combined procedure of 0 in these strata.

We consider various odds ratio estimates comparing the combined to the staged procedures with respect to the occurrence of complications; we did not include other covariates in these models or in deriving weights (see below), because other variables did not appear to confound the association of procedure type with outcome (Berlin et al. 1999); thus $V = \Phi$, the null set. Marginal models using a working independence correlation matrix show that combined procedure is associated with a substantially increased risk of complication (odds ratio (OR): 1.57, 95% confidence interval (CI) .99 – 2.48); the exchangeable working correlation matrix yields estimates closer to the null (OR: 1.18 CI: .81 – 1.73). A mixed effects model yields similar estimates (OR: 1.15 CI: .86 – 1.54). Conditional logistic regression, however, which controls for center, shows practically no association (OR: 1.02 CI: .75 – 1.37).

We fit a MSM using as weights the inverse of the center-specific probability of receiving the treatment level actually received. Like the conditional logistic regression, the MSM shows little effect of treatment (OR: 1.05, CI: .72 – 1.54). Unlike the conditional logistic regression, the MSM allows estimation of

the probability of success under either procedure. In the estimation, we used weighted generalized estimating equations (Hernan et al. 2000) with a working independence correlation matrix; our Web site (<http://www.cceb.upenn.edu/main/people/joffe.html>) provides Stata code used in this example.

The divergence among the estimates results from the divergence between the ecologic or between-center level association of staged procedures and complications and the within-center association. This association is visible in Figure 4, which plots the center-specific risk of complication against the center-specific probability of the combined procedure, along with a weighted (by the number of subjects at each center) least-squares fit.

It is difficult to provide reassurance that the conclusion from the MSM is correct, because this depends on an untestable assumption (3). Under (3), the type of procedure seems to have no effect in these data; the substantial differences among centers with respect to risk of complication and frequency of the combined procedures may stem from unobserved differences in patient mix or overall quality of care. Overall, this seems a reasonable explanation of the data.

In contrast to the MSM, the GEE estimate using the independence working correlation structure is reasonable if the choice of procedure is not associated with the risk of potential complications under either procedure; that is,

$$\Pr(A = 1|V, Y) = \Pr(A = 1|V), \quad (4)$$

where, unlike (3), independence does not condition on covariates like center that are part of X but not V . For (4) to be true and risk of complication to be unassociated with procedure choice

Table 4. Odds Ratios From Various Weighted and Unweighted Logistic Regression Models for the Association of Heparin With Surgical Success

Covariates in structural model (V)	Standard models (no weights)			Marginal structural models (weights)		
	Point estimate	95% confidence interval		Point estimate	95% confidence interval	
1. None	1.7	1.3	2.2	2.4	1.7	3.4
2. Age (3 terms), dialysis dependent, history of cardiovascular disease, prior vascular access, admission mean arterial pressure	2.2	1.6	3.0	2.9	2.1	4.0
3. Variables in 2, upstream venous diameter (three terms)	2.7	2.1	3.6	2.9	2.1	4.1

NOTE: Heparin use is classified as high (at least 1,000 units during surgery) or low (less than 1,000 units; includes no use).

overall, the higher risk of complication in centers with more combined procedures (Figure 4) must be offset by a lower risk of complication within center among subjects receiving the combined procedure (i.e., $E(Y^a|\mathbf{X}, A = 1) < E(Y^a|\mathbf{X}, A = 0)$).

To justify causal interpretation of estimates from the mixed effects or GEE exchangeable models, we need to assume in addition to (3) that, if cluster is associated with risk (i.e., $E(Y|A, \mathbf{X}, \mathbf{V}) \neq E(Y|A, \mathbf{V})$), it is not associated with treatment (i.e., $\Pr(A = 1|\mathbf{X}, \mathbf{V}) = \Pr(A = 1|\mathbf{V})$) (Neuhaus and Kalbfleisch 1998). Such assumptions are empirically testable; in our data, they may be rejected (see Figure 4). Unlike these methods, MSM estimates have causal interpretation under weaker assumptions, and an analysis of sensitivity to departures from these assumptions is available (Brumback, Hernan, Haneuse, and Robins 2004).

5. DISCUSSION

MSMs were originally proposed for controlling confounding by time-varying covariates when one is interested in estimating the joint effects of treatments or exposures received over a sustained period of time. Nonetheless, they can be useful in the simpler setting of scalar exposures or treatments, such as the examples presented here.

Confounding by measured factors results from imbalances in the distribution of predictors of outcome between groups defined by different levels of the observed treatment or exposure; for example, if $E(Y^a|\mathbf{X}) \neq E(Y^a)$ and $f(\mathbf{X}|A) \neq f(\mathbf{X})$. There are several ways to control for confounding, including weighting and statistical modeling; both of these approaches combine information about treatment effects from different strata of covariates $\mathbf{X}$. Weighting eliminates the imbalance in covariates $\mathbf{X}$ across groups. With statistical modeling, information about treatment effects (in generalized linear models) from different strata is combined through the use of modeling assumptions; for example, under a generalized linear model with no interactions between covariates and treatment, $E(Y|\mathbf{X}, A) = q(\mathbf{X}, A) = g^{-1}(\mathbf{X}\beta_X + A\beta_A)$, the assumption of a common treatment effect β_A is exploited.

Marginal structural models and their associated estimation methods are productively viewed as unifying these two approaches for controlling confounding. Solely using weighting is equivalent to letting $\mathbf{V}$ be the null set; standard modeling approaches require including all confounders in $\mathbf{X}$ in the $\mathbf{V}$ or the structural model. An alternate set of “stabilized” weights (Hernan et al. 2000), $\Pr(A|\mathbf{V})/\Pr(A|\mathbf{X})$, reduce to the standard equal weights when all confounders $\mathbf{X}$ are included in the structural model through $\mathbf{V}$, and so standard modeling estimates are also MSM estimates. MSMs and the associated weighted estimation methods allow more flexibility in the choice of variables to include in the structural model, since confounding may be controlled solely through weighting and so confounders need not be part of the structural model. This allows the structural model to reflect more closely the scientific or policy question at hand. The advantages of this are most apparent when one is interested in predicting a subject’s outcome under a given treatment but does not want to include certain confounding variables in the structural model.

MSMs are not always the ideal solution for controlling confounding, and suffer from the same problems as other weighted estimation approaches. When the true probability of receiving a given treatment level in some subset of the data is 0, assumption (3) is not satisfied; this can lead to bias. Similarly, estimated probabilities of 0 or 1 can also lead to bias. When the estimated probability is close to 0 in some subset of the data, some subjects (those who receive a treatment level which is very uncommon among subjects with the same values of covariates $\mathbf{X}$) will have very large weights. Large and variable weights can lead to unstable estimates; this may be mitigated by using “stabilized” weights (Hernan et al. 2000).

In multicenter studies with small centers, the estimated probability of treatment in some centers will be 0 or 1; for this reason, we eliminated the two smallest centers in the PTCA study. This may be appropriate if the number of subjects in those clusters is small relative to the total study, as in our example. Other options in this setting would be to pool centers with similar risks $E(Y^a|\mathbf{X})$ or to exclude small centers when designing the study. In contrast to MSMs, conditional likelihood methods deal appropriately with small centers. Further, continuous treatments or exposures will need to be categorized for MSMs, as we have done in the AVF example, again in contrast to conditional likelihood methods.

MSMs have some advantages over conditional likelihood methods. First, MSMs allow prediction, and estimates of marginal expectations or probabilities derive directly from regression estimates. Unlike MSMs, conditional logistic regression essentially ignores centers in which there are only successes or failures, and it also does not allow estimation of associations involving cluster-level or between-cluster covariates.

Both MSMs and conditional logistic regression provide ways to estimate parameters of interest while not estimating the association of nuisance variables with the outcome. MSMs can deal appropriately with this when the nuisance variables W (i.e., that part of $\mathbf{X}$ not in $\mathbf{V}$, the variables we wish to include in the model) are of high dimension. In contrast, in conditional logistic regression, a stratum will only be informative if it has at least two subjects discordant on $\mathbf{V}$ or A ; thus, when W is of high dimension or has continuous components and so almost all strata have only one observation, conditional logistic regression will not be appropriate.

In conclusion, MSMs and their associated estimation methods provide a useful addition to the methods available for controlling confounding. Further, the models help disentangle controlling confounding conceptually from variable selection.

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